Inorganic Cume

April-3-2015

Part-I. 2D-materials:

1. Find a pure elemental that can form a structure isoelectric to that of graphite (10').

2. Graphene, like a single graphite-sheet that is regarded as the "carbon in two-dimension", has been a very hot topic in today's materials science. Predict, with your reasoning, which chemical solvent you would recommend to use for dissolving or dispersing the graphene (10').

3. Many physicists and engineers predicted that the graphene will enable us to replace carbon nanotube in nanodevice making. But, experiences remind us that not all types of semiconductor could be easily coated on the graphene surface. Predict which semiconductor(s) would be very likely grown on the graphene, and give your reason(s) based on your chemistry knowledge. (20')

Part-II. 1D Materials

4. If the TiO_2-based nanotube's surface would be approximately the same as a typical TiO_2 bulk crystal surface in atomic scale, how would you functionalize the surface in order to make the nanotube surface to be hydrophobic and hydrophilic, respectively? (20')

5. (3). It is known that TiO_2-nanoparticles are the major ingredient in sunscreen lotion. Predict advantage(s) and disadvantage(s) in the development of a new sunscreen using the TiO2-nanotube as the major ingredient. (20')

6. (4). How would you use the TiO_2-nanotubes to study H_2-storage, drug release, and water photolysis? Please use examples or simple sketches to briefly discuss your chemistries in each of the three applications. (20')